

AMBIKA PRASAD MISHRA

RAILWAY SIGNALLING ENGINEER



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PROFILE

Signalling & Railway Systems Engineer with 7+ years of experience in safety-critical railway infrastructure, including tenure with Siemens and current involvement in KAVACH (TCAS) implementation under PMC at Egis India.

Specialized in Electronic Interlocking (EI), system integration, testing & commissioning, and fault diagnostics across multi-station railway environments. Proven ability to enhance system reliability, reduce failures, and ensure compliance with stringent railway safety standards.

Strong exposure to international signalling principles aligned with SIL-based safety systems, IEC standards, and ETCS/KAVACH frameworks, with hands-on experience in industrial communication protocols and embedded systems.

EXPERIENCE

Signalling Site Engineer (PMS KAVACH Project) | Egis India, April , 2026 – Present

- Managing PMS activities for Railway KAVACH projects, ensuring timely delivery, quality execution, and compliance with ETCS L2, SIL-4, and CENELEC safety standards (EN 50126/50128/50129).
- Coordinating with Indian Railways, contractors, and OEM teams for installation, integration, testing, and commissioning of TCAS and signaling systems.
- Preparing and maintaining DPRs, FAT/SAT reports, inspection records, commissioning documents, technical checklists, and progress logs for audit and regulatory compliance.
- Reviewing signaling layouts, EI interface circuits, network communication systems, and KAVACH architecture while resolving on-site technical issues with minimal project delays.
- Supporting FAT, SAT, integrated testing, and validation activities, ensuring adherence to fail-safe signaling principles, railway safety procedures, and operational reliability standards.

Signalling Troubleshooting Engineer | SIEMENS , April, 2022 – March, 2026

- Successfully installed, tested, and commissioned Electronic Interlocking (EI) systems across 8 railway stations in compliance with RDSO specifications, IRS signaling standards, and SIL-4 fail-safe safety requirements.
- Executed FAT, SAT, and CFAT activities with OEMs and Indian Railway representatives, ensuring EI system validation, reliability, and operational safety compliance.
- Diagnosed and resolved faults in PM, ROM, PIM, Embedded PCs, and EI subsystems, improving signaling system reliability and operational efficiency by 60%.
- Reduced system downtime by 80% through troubleshooting EI logic failures, checksum mismatches, relay interface issues, and SMB/network communication faults.
- Executed DMNJ, KPRR and PFU (ECOR) EI modification projects including SIP, RCC, TOC, relay testing, spare calculation, and interface testing using Delfos, ICS, GCIM, and GCSS tools.

Installation And Commissioning Engineer (GPS and Fuel Sensor)| i Teade Telematics, January , 2019 – March, 2021

- Experienced in GPS tracking system installation, testing, commissioning, and maintenance following automotive electrical safety and ISO quality standards.
- Skilled in calibration of capacitive and ultrasonic fuel sensors integrated with telematics and fleet management systems.
- Hands-on expertise in CAN Bus, RS232/RS485 communication, GSM/GPRS configuration, and electrical troubleshooting.
- Performed onsite commissioning, preventive maintenance, and fault diagnosis for vehicle monitoring systems.
- Managed technical documentation, customer coordination, and compliance with safety and quality procedures.

CORE TECHNICAL EXPERTISE

- Railway Signalling Systems: Electronic Interlocking (EI), KAVACH (TCAS), ETCS L2 Concepts, MSDAC
 - Testing & Commissioning: FAT, SAT, CFAT, SIT, System Validation & Safety Acceptance
 - Safety & Compliance: RDSO Standards, IEC 61508, EN 50126/28/29 (SIL Concepts)
 - Fault Diagnostics: Relay Logic, Interface Circuits, Root Cause Analysis (RCA), Fail-safe Systems
 - Communication Systems: Profinet, Profibus, Modbus, CAN, Ethernet, MVB
 - Embedded & Interfaces: UART, SPI, I2C, Industrial Networking
 - Tools: Siemens TIA Portal, Codesys, Delfos, ICS, GCIM, GCSS, Primavera, MPLAB X
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INTERNSHIP / TRAININGS

PLC, SCADA - Technical Trainee Intern | Chanakya Education Foundation | April, 2021 - March, 2022

- Gained hands-on experience in PLC, SCADA, HMI, and VFD systems for industrial automation applications.
- Developed automation logic improving production efficiency and operational reliability.
- Performed troubleshooting, commissioning, and maintenance for effective process control operations.

Embedded Systems - Training Program | Emertxe | July, 2025 - April, 2026

- Learned C, C++, data structures, and Linux internals for embedded software development and system-level programming applications.
 - Gained hands-on experience with microcontrollers, embedded C, memory management, interrupts, and hardware interfacing techniques.
 - Developed and tested embedded applications involving sensors, communication protocols, debugging, and hardware-software integration for reliable system performance.
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EDUCATIONS

B.Tech – Electrical and Electronics Engineering

Indotech College Of Engineering [August 2018 - July 2021]

Diploma – Electrical Engineering

Gandhi School Of Engineering [August 2015 - May 2017]

PORTFOLIO/KEY PROJECTS

TCAS KAVACH Implementation Project | Egis India, April 2026 – Present

- Contributed to ₹325 Cr railway KAVACH ETCS level-2 Implementation project from HWH to KGP focusing on installation, integration, and validation of signalling systems.
- Ensured compliance with signalling safety principles, system verification, and documentation processes.

Electronic Interlocking troubleshooting, testing and commissioning | SIEMENS , April 2022 – March 2026

- 32 Cr AMC contract of Indian Railway, ECOR for DMNJ, PFU and KPRR with 5 more main line stations under Waltair division and Khordha division.
- SIEMENS WESTRACE MK1 and MK2 Electronic interlocking : AMC, troubleshooting, testing and commissioning.

PLC based train accident avoidance system| Chanakya Education, August, 2021 – October, 2021

- Developed a PLC-based Train Accident Avoidance System using sensors, relay logic, and automated braking to prevent collisions and improve railway safety.
- Implemented ladder logic, HMI monitoring, and fail-safe interlocking for reliable train operation and signal coordination.